

Game System - PET

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1. Personality Questionnaire

General Disposition: My pet is a small owl that is naturally calm and a bit shy. Even though she is reserved, she becomes curious and slightly playful when she feels comfortable. As her energy decreases, her calm side becomes even more visible: she slows down, blinks more slowly, and looks for a safe place to rest.

Social Interaction: The owl is naturally affectionate. She reacts positively to attention and food, accepting almost anything the player offers. This makes her bond with the player very quickly and easily.

Adversity: When her energy is low, she becomes slow, tired, and less responsive. Her usual active night cycle feels heavy and difficult until she can rest again. When she is hungry, she tries to subtly call for attention by lightly flapping her wings, looking upward with big eyes, as if asking silently for help.

Success: When the player takes good care of her, the owl becomes more active. She flaps her wings faster and moves gently from side to side, showing excitement and satisfaction.

Unique Quirk: The owl has a characteristic and easy-to-implement habit: whenever she becomes curious, she tilts her head to one side and holds the pose for a few seconds. When relaxed, she stretches her wings slowly, similar to a gentle morning stretch.

Visual Design Impact: The personality directly dictates the animation blending. When she is **sleepy**, her movements slow down significantly, and her eyelids droop heavy and closed. When **hungry**, she becomes lethargic and moves her head rhythmically as if searching. In contrast, when she is **happy**, the visual feedback is immediate: she flies faster and displays much more expression in her wing movements.

Intended Player Experience: The game aims to evoke a sense of **calm** and **nurturing**. The player should feel a genuine bond of responsibility while interacting with the environment. There is a charming and slightly humorous contrast intended in the day/night cycle: seeing the owl wide awake and playful during the dark night, but struggling to stay awake as the sun rises, reinforcing the unique feeling of caring for a nocturnal creature.

2. State Machine Description

The system uses three numerical parameters, each ranging from 0 to 100 and updated continuously based on player interaction and time passage:

- **Hunger:** 100 means full; 0 means extremely hungry.
- **Energy:** 100 means fully rested; 0 means exhausted.
- **Confidence:** Controls the mood from shy to happy.

The FSM has three main Blend Trees:

IdleBlend (Idle ↔ Happy)

Controlled by Confidence:

- 0 = Idle
- 100 = Happy

EatBlend (AskForEating ↔ Eating)

Controlled by Hunger:

- 0 = AskForEating
- 40 = Eating

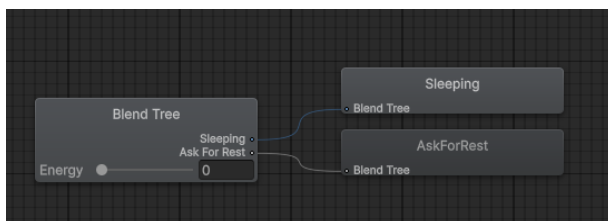
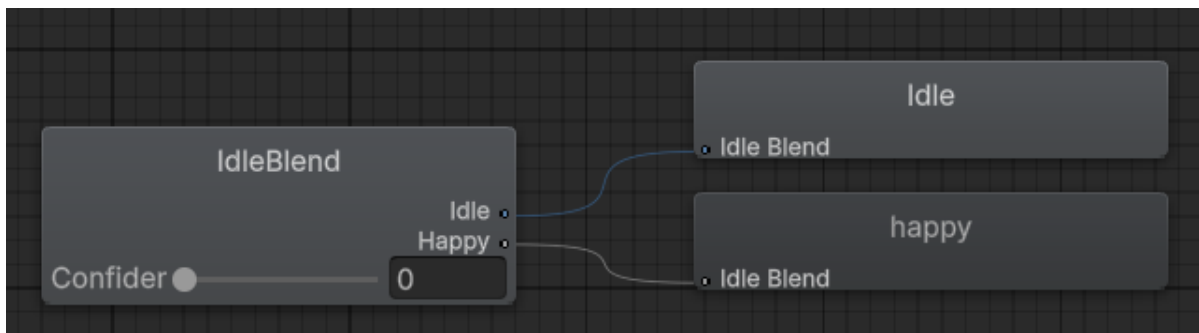
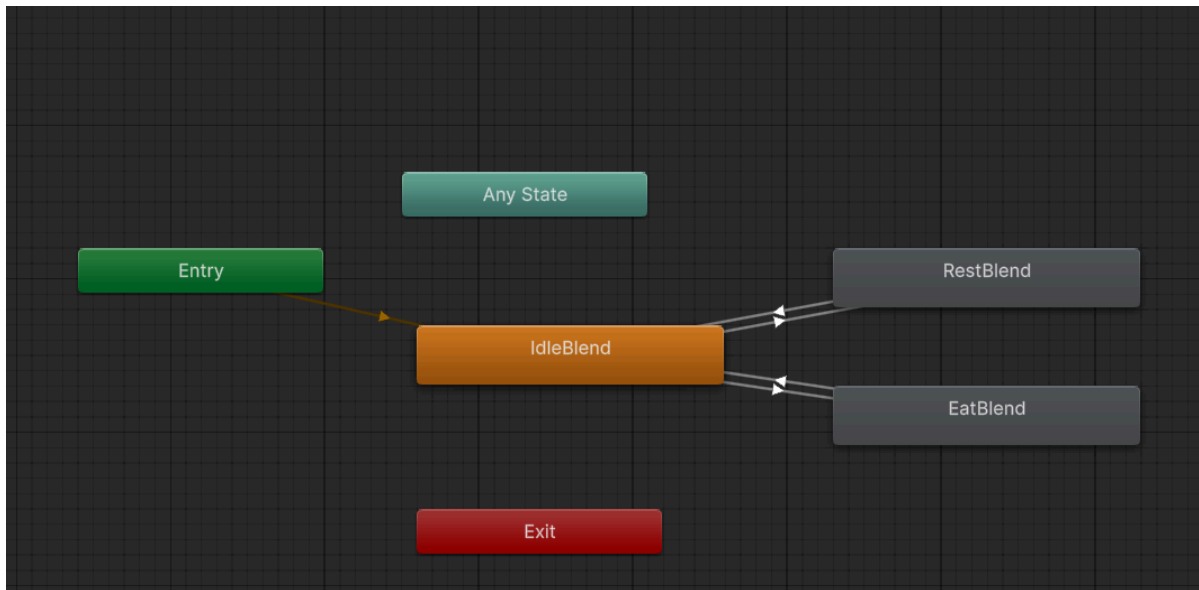
RestBlend (AskForRest ↔ Sleeping)

Controlled by Energy:

- 0 = Sleeping
- 30 = AskForRest

State Transitions:

- IdleBlend → EatBlend when Hunger < 40
- EatBlend → IdleBlend when Hunger > 70
- IdleBlend → RestBlend when Energy < 30
- RestBlend → IdleBlend when Energy > 60
- *Note: IdleBlend stays active for mood changes; Confidence controls Idle↔Happy inside the Blend Tree.*



3. UI and Player Interaction

The UI uses two main buttons:

- **Feed Button:** Increases Hunger, increases Confidence, and may slightly reduce Energy.
- **Sleep Button:** Increases Energy, slightly reduces Hunger, and may increase Confidence.

Each button updates the internal stats and then synchronizes them with the Animator using:

C#

```
animator.SetFloat("Hunger", hunger);  
animator.SetFloat("Energy", energy);  
animator.SetFloat("Confidence", confidence);
```

The continuous update loop reduces Hunger over time, reduces Energy while awake, increases Energy during rest, and slowly adjusts Confidence based on player behavior.



4. Playtesting & Evaluation

The prototype was playtested by 10 participants, each interacting with the owl for approximately 2–5 minutes before answering a short questionnaire. The goal of the playtest was to evaluate the clarity of the interaction, the readability of the animations, and the emotional impact of the pet.

The main questions used in the playtest were:

- *How much time did you spend testing the owl?*
- *Could you understand how to let the owl rest/sleep?*
- *Could you differentiate the owl's animations when she was hungry, tired, or happy?*
- *Did the animations match the owl's internal state (Hunger, Energy, Confidence)?*
- *How did you feel while interacting with the owl?*
- *Did you enjoy the day/night cycle?*
- *What did you like the most about the experience?*

Playtest Results

Most players spent between **2 and 5 minutes** interacting with the owl, which was sufficient for them to observe multiple state changes.

All participants reported that it was **easy to understand how to let the owl rest or sleep**, selecting “Yes, *easily*”. This indicates that the core interaction design is clear and accessible.

The majority of players were able to **differentiate the owl's animations** across multiple states such as *Hungry, Asking for food, Tired, Asking for rest, Sleeping, and Happy*. This confirms that the non-verbal communication through animation and Blend Trees is effective.

When asked whether the animations matched the internal states (Hunger, Energy, Confidence), most players answered “Yes, *clearly*” or “Yes, *somewhat*”. Only a small minority reported uncertainty, suggesting that the system logic is largely readable from visual feedback alone.

Emotionally, the feedback strongly aligned with the intended experience. Players described feeling **relaxed, calm, and responsible** for the owl:

“I felt relaxed. The owl is very charismatic and her reactions made me feel like I was actually taking care of a pet.”

“I felt responsible for the owl. When it looked tired or hungry, I wanted to help it immediately.”

The **day/night cycle** was frequently mentioned as one of the most enjoyable aspects of the experience. Several players explicitly highlighted it as their favorite feature:

“The day and night cycle combined with the animations was my favorite part. It helped me understand when the owl was tired and made the world feel alive.”

Some testers suggested small improvements, such as adding more functions or making the connection between the day/night cycle and the owl's tiredness even clearer. These comments indicate interest in deeper interaction rather than confusion, and provide a clear direction for iteration.

Design Iteration Based on Feedback

Based on the playtest data, the following improvements are planned:

- Strengthen the visual link between the day/night cycle and the owl's energy level.
- Add minor onboarding cues for first-time players, explaining the feed and sleep buttons.
- Introduce additional small reactions or interactions to deepen engagement.

The playtest results confirm that the system is emotionally engaging, readable, and aligned with the intended experience of caring for a gentle nocturnal companion.

[Google Forms Link](#)

[Game Link](#)

5. Summary

The combination of Blend Trees, numerical parameters, and state transitions creates a responsive virtual pet capable of expressing hunger, tiredness, curiosity, happiness, and subtle emotions through smooth non-verbal animations. The system is simple to maintain, expressive, and closely aligned with the personality design of the owl.